Felix Pultar, PhD

PERSONAL INFORMATION

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Website: `https://pultar.ai`

PROFILE

Felix Pultar is a Senior Research Scientist at Microsoft Research AI for Science based in Amsterdam, the Netherlands. His research focuses on AI-driven approaches to accelerate synthetic organic chemistry and molecular design, including AI co-scientists and autonomous agents for scientific discovery. His representative works include the retrosynthesis prediction model RetroChimera and the neural network potential AMP. Before joining Microsoft, he was a postdoctoral researcher at ETH Zurich with Sereina Riniker, where he developed neural network potentials and equivariant graph neural networks for molecular simulation. He obtained his Ph.D. in synthetic organic chemistry with Erick M. Carreira at ETH Zurich.

PROFESSIONAL EXPERIENCE

Senior Research Scientist **Since 2026**
Microsoft Research, Netherlands
AI for Science

- Co-developing and evaluating AI models for retrosynthetic planning (RetroChimera) and chemical procedure generation (QFANG)
- Close research collaboration with GSK and other partners on drug discovery projects
- Pioneering scalable agent and AI co-scientist infrastructure

Senior Scientist and Postdoctoral Researcher **2021 – 2025**
ETH Zurich, Switzerland
Riniker Research Group

- Conceived, developed, and deployed neural networks based on a novel message-passing formalism (AMP) for the fast and accurate modeling of proteins and ligands
- Co-led end-to-end project lifecycle: PyTorch-based large-scale foundation model training and optimized C++/CUDA production inference on thousands of GPUs
- Planned, generated, and managed large DFT datasets (> 1.5 million systems)

Doctoral Researcher **2017 – 2021**
ETH Zurich, Switzerland
Carreira Research Group

- Total synthesis and retrosynthesis, structural elucidation, and biological investigations of complex natural products from the human microbiome

Research Intern **2015**
Bayer Pharmaceuticals, Wuppertal, Germany

SKILLS

Programming languages: Python, C++, Swift, Java, C, SQL, Bash, CUDA
Libraries: PyTorch, TensorFlow, sklearn, NumPy, SciPy, Pandas, OpenMP, MPI
Computational software: OpenMM, ORCA, GROMOS, RDKit, PyMOL, among others
Miscellaneous: DevOps & MLOps (Docker, Git, CI/CD), HPC and Azure
Languages: English, German, Spanish, Dutch

EDUCATION

Doctor of Sciences (Dr. sc. ETH Zurich, Chemistry) **2017 – 2021**
ETH Zurich, Switzerland
Recipient of the Scholarship Fund of the Swiss Chemical Industry

Bachelor & Master of Science (BSc. & MSc. Chemistry) **2010 – 2016**
University of Munich, Germany
Top 2 in cohort (best final exams, second-best overall grade average)
Exchange: University of Toronto, Canada, asymmetric transition metal catalysis

Publications

- Maziarz, K.†; Liu, G.†; Pultar, F.†; Gardner, J.; Gensch, T.; Helie, J.; Misztela, H.; Tripp, A.; Li, J.; Kornev, A.; Gaiński, P.; Hoefling, H.; Fortunato, M.; Gupta, R.; Baxter, A.; Poole, D. L.; Elward, J. M.; Krzyzanowski, A.; Pogány, P.; Pickett, S. D.; Wall, I. D.; Bishop, C. M.; Humphreys, P. G.; Lumley, J. A.; Wiesenfeldt, M. P.; Segler, M. H. S. Chemist-aligned retrosynthesis by ensembling diverse inductive bias models. *Nature* **2026**. https://doi.org/10.1038/s41586-026-11160-9.

- Thürlemann, M.†; Pultar, F.†; Gordiy, I.; Riniker, S. Biomolecular Multiscale Simulation (BMS25) Dataset to Train Neural Network Potentials for QM/MM Settings with Electrostatic Embedding. *Sci. Data.* **2026**. https://doi.org/10.1038/s41597-026-08138-7.

- Thürlemann, M.†; Pultar, F.†; Gordiy, I.†; Ruijsenaars, E.; Riniker, S. Multiscale Neural Network Potential with Anisotropic Message Passing for the Fast and Accurate Simulation of Protein Dynamics and Enzymatic Reactions. *J. Am. Chem. Soc.* **2026**, *148*, 28133–28156. https://doi.org/10.1021/jacs.6c00217.

- Kuhn, A. S.†; Gordiy, I.†; Pultar, F.†; Riniker, S. Leveraging the Potential of Machine-Learning Interatomic Potentials for QM/MM Simulations. *CHIMIA* **2026**, *80*, 298–304. https://doi.org/10.2533/chimia.2026.298.

- Brandl, P.†; Lüthy, L.†; Pultar, F.†; Hansen, M. E.; Jelmini, N.; Faris, J.; Yasmin, S. O.; Kamenik, A. S.; Arnold, R.; Ebert, M.-O.; Wolfrum, S.; Lokey, R. S.; Riniker, S.; Carreira, E. M. Mutanobactin D from the Human Microbiome: Chemistry, Biology, and Molecular Dynamics Studies. *J. Am. Chem. Soc.* **2025**, *147*, 43330–43341. https://doi.org/10.1021/jacs.5c08850.

- Katzberger, P.†; Pultar, F.†; Riniker, S. Transferring Knowledge from MM to QM: A Graph Neural Network-Based Implicit Solvent Model for Small Organic Molecules. *J. Chem. Theory Comput.* **2025**, *21*, 7450–7459. https://doi.org/10.1021/acs.jctc.5c00728.

- Pultar, F.†; Thürlemann, M.†; Gordiy, I.; Doloszeski, E.; Riniker, S. Neural Network Potential with Multiresolution Approach Enables Accurate Prediction of Reaction Free Energies in Solution. *J. Am. Chem. Soc.* **2025**, *147*, 6835–6856. https://doi.org/10.1021/jacs.4c17015.

- Poliak, P.; Bleiziffer, P.; Pultar, F.; Riniker, S.; Oostenbrink, C. A Robust and Versatile QM/MM Interface for Molecular Dynamics in GROMOS. *J. Comput. Chem.* **2025**, *46*, e70053. https://doi.org/10.1002/jcc.70053.

- Nikolic, S.; Alastra, G.; Pultar, F.; Lüthy, L.; Stadlinger, B.; Carreira, E. M.; Bugueno, I. M.; Mitsiadis, T. A. Mutanobactin-D, a *Streptococcus mutans* Non-Ribosomal Cyclic Lipopeptide, Induces Osteogenic/Odontogenic Differentiation of Human Dental Pulp Stem Cells and Human Bone Marrow Stem Cells. *Int. J. Mol. Sci.* **2025**, *26*, 1144. https://doi.org/10.3390/ijms26031144.

- Cotter, E.; Pultar, F.; Riniker, S.; Altmann, K.-H. Experimental and Theoretical Studies on the Reactions of Aliphatic Imines with Isocyanates. *Chem. Eur. J.* **2024**, *30*, e202304272. https://doi.org/10.1002/chem.202304272.

- Pultar, F. Total Synthesis and Biological Evaluation of Mutanobactin D from the Human Microbiome. Ph.D. Dissertation, ETH Zurich, **2021**. https://doi.org/10.3929/ethz-b-000505539.

- Pultar, F.; Hansen, M. E.; Wolfrum, S.; Böselt, L.; Fróis-Martins, R.; Bloch, S.; Kravina, A. G.; Pehlivanoglu, D.; Schäffer, C.; LeibundGut-Landmann, S.; Riniker, S.; Carreira, E. M. Mutanobactin D from the Human Microbiome: Total Synthesis, Configurational Assignment, and Biological Evaluation. *J. Am. Chem. Soc.* **2021**, *143*, 10389–10402. https://doi.org/10.1021/jacs.1c04825.

- Johnson, T.†; Pultar, F.†; Menke, F.; Lautens, M. Palladium-Catalyzed α-Arylation of Vinylogous Esters for the Synthesis of γ,γ-Disubstituted Cyclohexenones. *Org. Lett.* **2016**, *18*, 6488–6491. https://doi.org/10.1021/acs.orglett.6b03394.

- Tambornino, F.; Sappl, J.; Pultar, F.; Cong, T. M.; Hübner, S.; Giftthaler, T.; Hoch, C. Electrocrystallization: A Synthetic Method for Intermetallic Phases with Polar Metal–Metal Bonding. *Inorg. Chem.* **2016**, *55*, 11551–11559. https://doi.org/10.1021/acs.inorgchem.6b02068.

- Arantes, G. M.; Brown, R. O.; Crehuet, R.; Cui, Q.; Ensing, B.; Field, M. J.; Gordiy, I.; Lameira, J.; Marino, T.; Pultar, F.; Riniker, S.; Ryde, U.; Serapian, S. A.; Tabacchi, G. Best Practices for Hybrid QM/MM Simulations. *LiveCoMS, manuscript in preparation.*

† Authors contributed equally.

Talks and Posters

- **2026** 'Scaling Message Passing to Simulate Proteins with DFT Accuracy'. Data & ML Lunch, AI4Science club, ETH Zurich, Switzerland. Presentation.
- **2023** 'Enhancing Molecular Dynamics Simulations through Deep Learning'. Platform for Advanced Scientific Computing (PASC) conference, Davos, Switzerland. Presentation.
- **2023** 'Developments for QM/MM with GROMOS'. 30th Intl. BIOMOS Symposium, Mutters, Austria. Presentation.
- **2023** 'Enhancing QM/MM MD simulations: combination of semi-empirical methods with a Δ-learning scheme'. ACS Spring conference, Indianapolis, USA. Presentation.
- **2022** 'Enhancing QM/MM MD simulations: combination of semi-empirical methods with a Δ-learning scheme'. CECAM Workshop: Present and Future of Hybrid Quantum Chemical and Molecular Mechanical Simulations, Lecco, Italy. Presentation.
- **2022** 'Exploring Natural Products from the Human Microbiome: Total Synthesis and Configurational Assignment of Mutanobactin D'. Tetrahedron Symposium, Lisbon, Portugal. Poster presentation.
- **2022** 'Deciphering Interspecies Networking of Human Microbiota'. SSCI scholarship awardees company visit, Janssen AG, Schaffhausen, Switzerland. Presentation.
- **2021** 'Infrared Sequence Alignment Enables Stereochemical Assignment of Complex Natural Products'. 7th ORCA User Meeting, Webinar. Presentation.
- **2020** 'Synthetic Studies of Structurally Intriguing Metabolites from the Human Microbiome'. Symposium of the Scholarship Fund of the Swiss Chemical Industry, ETH Zürich, Switzerland. Presentation.
- **2014** 'New Building Blocks for Alkaloid Synthesis'. Undergraduate Research Conference on Molecular Sciences (URCUP), Wildbad Kreuth, Germany. Poster presentation.

Teaching Experience

Introduction to Computational Chemistry **2023**
ETH Zurich, Switzerland and Yale University, USA

- Co-designed and taught workshop on quantum chemical calculations for an audience of a hundred graduate students, postdocs, and faculty of organic chemistry

Teaching Assistant **2017 – 2024**
ETH Zurich, Switzerland

- Taught at the BSc. and MSc. levels of the chemistry and CS curricula; supervised four Master thesis students and trained a laboratory technician over three years